

# Unit 2 Guided Notes — Exponents & Scientific Notation

Same base? Move the exponents. Big or tiny number? Pack it into a  $\times 10^n$ .

Standards: NC.8.EE.1 · NC.8.EE.3 · NC.8.EE.4

 Blank Worksheet (PDF)

 Answer Key (PDF)

 Official PDF Packet

 EC/Modified Notes (PDF)

This packet has three parts: **Part 1 — Guided Notes** (fill in the blanks as you learn), **Part 2 — Practice Problems**, and the **Answer Key** built right in. Work the blanks on your printed copy first, then click any blank, step, or check button on screen to reveal the answer.

## 1 Exponent Rules

★ **KEY RULE:** Exponent rules only apply when the BASES are the SAME!

### Product Rule — Count the Factors

$$\begin{array}{c} x^3 \\ \boxed{x} \boxed{x} \boxed{x} \end{array} \times \begin{array}{c} x^2 \\ \boxed{x} \boxed{x} \end{array} = \begin{array}{c} \boxed{x} \boxed{x} \boxed{x} \\ \boxed{x} \boxed{x} \end{array} = x^5$$

$x^2 = x^{3+2} = x^5 \rightarrow$  count the factors:  $3 + 2 = 5 \rightarrow$  ADD the exponents

### The Five Core Rules

Fill the examples and key phrase on your printed copy, then click to check.

| Rule              | Formula                    | Simple Example        | Monomial Example             | Key Phrase               |
|-------------------|----------------------------|-----------------------|------------------------------|--------------------------|
| Product Rule      | $x^a \times x^b = x^{a+b}$ | $x^3 \cdot x^4 = x^7$ | $(2x^3)(3x^2) = 6x^5$        | Same base? ADD           |
| Quotient Rule     | $x^a \div x^b = x^{a-b}$   | $x^6 \div x^2 = x^4$  | $(12x^7) \div (3x^2) = 4x^5$ | Same base? SUBTRACT      |
| Power Rule        | $(x^a)^b = x^{a \times b}$ | $(x^2)^3 = x^6$       | $(2x^3)^2 = 4x^6$            | Power to power? MULTIPLY |
| Zero Exponent     | $x^0 = 1$                  | $5^0 = 1$             | $(3x^5)^0 = 1$               | ANY base <sup>0</sup> =1 |
| Negative Exponent | $x^{-n} = 1/x^n$           | $2^{-3} = 1/8$        | $4x^{-3} = 4/x^3$            | Flip to fraction         |

⚠ **PARENTHESES MATTER!**  $-3^2 = -9$  vs.  $(-3)^2 = 9$ . Without ( ), ONLY the base gets the exponent — not the negative sign!

## Modeling the Rules — See WHY

Each rule comes from writing out the repeated multiplication. Click each card to reveal the reasoning.

### Product Rule · $x^3 \times x^2 = x^5$

$x \cdot x \cdot x \times x \cdot x = x \cdot x \cdot x \cdot x \cdot x = x^5$ . Count the factors:  $3 + 2 = 5$ . Same base → ADD the exponents.

### Quotient Rule · $x^5 \div x^2 = x^3$

$x \cdot x \cdot x \cdot x \cdot x \div x \cdot x \rightarrow$  cancel 2 matching pairs  $\rightarrow x \cdot x \cdot x = x^3$ . Cancel matching factors:  $5 - 2 = 3$ . Same base → SUBTRACT the exponents.

### Power Rule · $(x^2)^3 = x^6$

Three copies of  $x^2$ :  $(x \cdot x)(x \cdot x)(x \cdot x) = x^6$ .  $2 \times 3 = 6$ . Power to a power → MULTIPLY.

### Zero Exponent · $x^0 = 1$

$x \cdot x \cdot x \div x \cdot x \cdot x = 1$  (anything divided by itself is 1) — but by the quotient rule this also equals  $x^{3-3} = x^0$ . So  $x^0 = 1$ .

### Negative Exponent · $x^{-3} = 1/x^3$

$x \cdot x \div x \cdot x \cdot x \cdot x \cdot x = x^{2-5} = x^{-3}$  — but cancelling factors leaves  $1/x^3$ . So a negative exponent means "flip it":  $x^{-n} = 1/x^n$ .

## Quick Check

**MC** Simplify  $x^5 \cdot x^3$ . Which rule applies, and what is the result?

✓ Same base, multiplication → Product Rule: ADD the exponents.  $x^5 \cdot x^3 = x^8$ .

## Match the Rule to its Key Phrase

**MATCH** Click a rule, then its key phrase.

### Rule

Product Rule

Quotient Rule

Power Rule

Negative Exponent

### Key Phrase

Power to power? MULTIPLY

Same base? ADD exponents

Flip to a fraction

Same base? SUBTRACT exponents

0 of 4 matched

### Worked Example — Simplify $(4x^3y^2)(3x^5y)$

Step 1 – Multiply coefficients:  $4 \times 3 = 12$

Step 2 – Add x exponents:  $x^3 \cdot x^5 = x^8$

Step 3 – Add y exponents:  $y^2 \cdot y^1 = y^3$

✓ Step 4 – Combine:  $12x^8y^3$

**ERROR ANALYSIS** Simplify  $(3x^2)^3$ . A student wrote  $3x^6$ . What went wrong?

The coefficient 3 must ALSO be cubed:  $3^3 = 27$ . Correct:  $27x^6$ . The power rule applies to EVERYTHING inside the parentheses.

## 2 Scientific Notation

★ **THE FORMAT:**  $a \times 10^n$ , where  $1 \leq a < 10$  — example:  $4.5 \times 10^7$  = coefficient · base (always 10) · exponent.

### Move the Decimal

45,000  $\xrightarrow{\text{move LEFT 4}}$   $4.5 \times 10^4$   
big number exponent +4

0.0067  $\xrightarrow{\text{move RIGHT 3}}$   $6.7 \times 10^{-3}$   
small number exponent -3

| Type                   | Direction          | Exponent | Example                       |
|------------------------|--------------------|----------|-------------------------------|
| Large number           | Move decimal LEFT  | POSITIVE | $45,000 = 4.5 \times 10^4$    |
| Small number ( $< 1$ ) | Move decimal RIGHT | NEGATIVE | $0.0067 = 6.7 \times 10^{-3}$ |

### Multiplying — 3-Step Process

- MULTIPLY** the coefficients (the a-values)
- ADD** the exponents
- Adjust if coefficient is NOT between **1** and **10**

**Example (no adjustment):**  $(3 \times 10^4) \times (2 \times 10^5)$

Multiply coefficients:  $3 \times 2 = 6$

ADD exponents:  $4 + 5 = 9$

✓ Answer:  $6 \times 10^9$

### Dividing — 3-Step Process

1. **DIVIDE** the coefficients
2. **SUBTRACT** the exponents
3. Adjust if needed

**Example (no adjustment):  $(8 \times 10^6) \div (4 \times 10^2)$**

Divide coefficients:  $8 \div 4 = 2$

SUBTRACT exponents:  $6 - 2 = 4$

✓ Answer:  $2 \times 10^4$

⚠ **ADJUSTMENT RULES:** If coefficient  $\geq 10$ : move decimal LEFT, ADD 1 to exponent. If coefficient  $< 1$ : move decimal RIGHT, SUBTRACT 1 from exponent.

**Example needing adjustment:  $(5 \times 10^3) \times (4 \times 10^2)$**

Multiply:  $5 \times 4 = 20$

Add exponents:  $3 + 2 = 5 \rightarrow$  result:  $20 \times 10^5$

Adjust ( $20 \geq 10$ ):  $20 = 2.0 \times 10^1$ , so ADD 1 to the exponent

✓ Final answer:  $2.0 \times 10^6$

**ERROR ANALYSIS** Convert 0.0045 to scientific notation. A student wrote  $4.5 \times 10^3$ . What's the mistake?

Small number  $\rightarrow$  NEGATIVE exponent. Correct:  $4.5 \times 10^{-3}$ . Numbers less than 1 always have negative exponents in scientific notation.

➡ Go to Unit 2 Practice  $\rightarrow$

## Part 2 — Practice Problems

✦ **Show all work on your printed copy.** Circle your final answer, then check on screen.

### A. Exponent Rules with Monomials

1) Simplify  $x^3 \cdot x^5$

answer

**2) Simplify  $(y^4)^3$**

answer

**3) Simplify  $a^8 \div a^3$**

answer

**4) Simplify  $(2x^3)(5x^2)$**

answer

**5) Simplify  $(3m^2)^3$**

answer

**6) Simplify  $12y^7 \div 4y^2$**

answer

**7) Simplify  $(-2a^4)^2$**

answer

**8) Simplify  $(5x^2y)(3xy^3)$**

answer

**9) Simplify  $18n^6 \div 6n^6$**

answer

**10) Simplify  $(4b^3)^2 \div 2b$**

answer

**11) Simplify  $(-3x^2)(-4x^3)$**

answer

**12) Simplify  $(x^5y^2)^3$**

answer

## **B. Scientific Notation Conversions**

**1) 45,000,000 → scientific notation**

answer

**2) 0.00032 → scientific notation**

answer

**3)  $6.02 \times 10^8$  → standard form**

answer

**4)  $9.1 \times 10^{-5}$  → standard form**

answer

**5) 780,000 → scientific notation**

answer

**6) 0.0000067 → scientific notation**

answer

**7)  $3.45 \times 10^4$  → standard form**

answer

8)  $1.2 \times 10^{-3} \rightarrow$  standard form

9) 5,280,000,000  $\rightarrow$  scientific notation

10)  $2.5 \times 10^{-2} \rightarrow$  standard form

### C. Operations

1)  $(3 \times 10^4)(2 \times 10^5)$

2)  $(8 \times 10^7) \div (4 \times 10^3)$

3)  $(5 \times 10^3)(7 \times 10^6)$

4)  $(9.6 \times 10^8) \div (3.2 \times 10^5)$

### D. Word Problems

1. The Sun is about  $9.3 \times 10^7$  miles from Earth. Light travels  $1.86 \times 10^5$  miles per second. How many seconds does it take light to reach Earth?

$(9.3 \times 10^7) \div (1.86 \times 10^5) = 5 \times 10^2$  seconds (500 seconds).

2. A bacteria cell is about  $2 \times 10^{-6}$  m long. A virus is about  $5 \times 10^{-8}$  m long. How many times longer is the bacteria?

$(2 \times 10^{-6}) \div (5 \times 10^{-8}) = 0.4 \times 10^2 = 40$  times longer.

**3. The US national debt is about  $3.4 \times 10^{13}$  dollars. The US population is about  $3.4 \times 10^8$  people. What is the debt per person?**

$$(3.4 \times 10^{13}) \div (3.4 \times 10^8) = 1 \times 10^5 = \$100,000 \text{ per person.}$$

**4. A grain of sand has mass  $6.5 \times 10^{-4}$  grams. A beach has about  $8 \times 10^{18}$  grains of sand. What is the total mass in grams?**

$$(6.5 \times 10^{-4})(8 \times 10^{18}) = 52 \times 10^{14} = 5.2 \times 10^{15} \text{ grams.}$$

**5. There are about  $3.7 \times 10^{13}$  cells in a human body. Each cell is about  $1 \times 10^{-5}$  m wide. If all cells were lined up, how long in meters?**

$$(3.7 \times 10^{13})(1 \times 10^{-5}) = 3.7 \times 10^8 \text{ meters.}$$

**6. A computer does  $4.8 \times 10^9$  calculations per second. A task needs  $1.44 \times 10^{12}$  calculations. How many seconds to finish?**

$$(1.44 \times 10^{12}) \div (4.8 \times 10^9) = 0.3 \times 10^3 = 3 \times 10^2 \text{ seconds (300 seconds).}$$

[← Back to Unit 2](#)